

ZAALI Mohamed

AI Software Engineer

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EDUCATION

ENSAM Rabat - State Engineer Student in Data Science, AI & Digital Health

Sep 2023–Present

PROFESSIONAL EXPERIENCE

Syntax - Casablanca - AI Agent Developer Intern

Feb 2026 - Present

Building an extensible MCP server connecting AI agents to Microsoft 365 via Microsoft Graph API.

- Reduced agent configuration complexity by **architecting a preset system** that ships context-aware tool sets.
- Built a **5-step pipeline** that auto-generates typed MCP tools from Microsoft's OpenAPI spec.
- Enabled any M365 service to be onboarded with **zero code changes** by designing a JSON-driven extensibility architecture.

Al Barid Bank - Rabat - Full Stack Developer Intern

Jul 2025 - Sep 2025

Design and development of a platform for analyzing POS, SoftPOS, and e-commerce transactions.

- Accelerated transaction data access by building a **secure REST API** serving aggregated POS, SoftPOS, and e-commerce metrics.
- Empowered analysts to explore trends by delivering a **responsive dashboard** with interactive charts and drill-down navigation.
- Ensured long-term maintainability by designing the full **UML software architecture** before implementation, reducing rework risk.

Ministry of Youth, Culture and Communication - Rabat - Data Science Intern

Jul 2024 - Sep 2024

Development of a predictive analytics system for cultural heritage preservation.

- Enabled proactive preservation decisions by building a **predictive model** forecasting cultural heritage degradation risk.
- Made insights accessible to non-technical experts by integrating model predictions into an interactive **Flask web application**.
- Ensured production availability by deploying the application on an **Ubuntu server** with Unicorn and Nginx.

PROJECTS & ACCOMPLISHMENTS

- Project Architect Agent - Autonomous Project Documentation Agent:** Agentic system that takes a project idea, runs a **research loop** per implementation step, and generates a structured **Notion workspace**. | Technologies: LangGraph OpenAI API Gradio Notion API
- AI Invoice Processing Agent - Automated Invoice Approval Workflow:** Agent that extracts, validates, and approves PDF invoices with business rule enforcement and email notifications. | Technologies: LangGraph Gradio OpenAI API Gmail API
- Twitter MCP Server - Tool for AI Agent Social Media Automation:** Exposes Twitter/X tools (post, thread, search) to any MCP-compliant AI agent via Streamable HTTP and SSE endpoints. | Technologies: FastMCP Python Twitter API
- El-Qess Restaurant AI Assistant - Langflow-Powered Restaurant Agent:** Conversational agent handling **FAQ via RAG**, table reservations, and confirmation emails. | Technologies: LangFlow OpenAI API Chroma DB Google Calendar MCP Gmail MCP
- AL-MUQARRIR - Automatic Transcription & Meeting Minutes Generation:** Web platform for **real-time transcription**, speaker **diarization**, and automatic generation of **meeting minutes** with **semantic analysis**. | Technologies: Whisper PyAnnote Gemini Flask Angular
- Cura - Disease & Medication Tracking:** Web application for treatment management with AI-powered **drug interaction detection** and generation of patient-friendly alerts. | Technologies: PyTorch Gemini API Flask SQLAlchemy Bootstrap SQLite
- LipReader - Lip-Reading Application:** Application converting lip movements into text. | Technologies: TensorFlow Streamlit
- Real-Time Network Intrusion Detection System:** System capturing and analyzing network packets in real-time. | Technologies: Python Kafka PyShark Pandas Scikit-learn Joblib

TECHNICAL SKILLS

AI Engineering: LangGraph LangChain LangFlow FastMCP OpenAI API Gemini API RAG

Programming Languages: Python Java TypeScript JavaScript

ML/DL: PyTorch TensorFlow Scikit-learn NLP

Databases: PostgreSQL MySQL MongoDB SQLite

Web Tech & Frameworks: HTML CSS Bootstrap Angular Flask FastAPI Spring Boot Streamlit Gradio

EXTRACURRICULAR ACTIVITIES

Learning Experience Architect - Fatal Error Club ENSAM Rabat

SEP 2024 - JUN 2025

- Led workshops in data science, machine learning, and software development; co-created innovative programs.
- Optimized training outcomes through continuous evaluation and refinement.

CERTIFICATIONS

• CCNA: Introduction to Networks

• Data Structures & Algorithms in Python

• Python Programming Fundamentals

• SQL Fundamentals

• Introduction to Data

LANGUAGES

• English: Fluent

• French: Fluent

• Arabic: Native